

Spring 2026 – Claudeversion#2

Noetic Learning Mathematics Contest

Grade 3

45 minutes • No calculators allowed

Student Name: _____

For Teacher Use Only

Number of correct answers: _____

Score (Number of correct answers $\times$ 5): _____

Instructions: Write your answer on the answer line. Use the shaded Work Space box beside each question for scratch work. No calculators. Good luck!

Questions

Work Space

1) $156 - 79 = ?$

Answer: _____

Work space

2) Jack has 3 boxes of crayons. Each box holds 12 crayons. He gives 7 crayons to a classmate. How many crayons does Jack have left?

Answer: _____ crayons

Work space

3) A pattern starts: 3, 7, 11, 15, ... (add 4 each time). What is the 6th number?

Answer: _____

Work space

4) Five friends share 45 grapes equally. Each friend then eats 3. How many grapes does each friend have left?

Answer: _____ grapes

Work space

5) Ride tickets cost \$4 each. Sam has \$30 and buys 3 tickets. How much money does Sam have left?

Answer: _____ dollars

Work space

6) A square tile has a side length of 9 cm. What is the perimeter of the tile?

Answer: _____ centimeters

Work space

7) Today is Thursday. A field trip is in 20 days. What day of the week will the field trip be?

Answer: _____

Work space

8) A library shelf has 120 books. Students check out 35, then 18 are returned, then 22 more are checked out. How many books are on the shelf now?

Answer: _____ books

Work space

9) 30 students: $\frac{1}{5}$ play piano, $\frac{1}{3}$ play soccer (no overlap). How many students play piano or soccer?

Answer: _____ students

Work space

10) Think of a number. Double it, then add 15. The result is 47. What is the number?

Answer: _____

Work space

Questions

Work Space

11) A train departs at 9:45 AM. The ride takes 2 hours 40 minutes. What time does it arrive?

Answer: _____ AM / PM

Work space

12) Alex=6, Beth=4, Dana=8, Evan=2. Each holds two identical cards. All 10 cards sum to 50. What number does Cole hold on each card?

Answer: _____

Work space

13) A printer prints 8 pages per minute. It runs from 2:10 PM to 2:25 PM. How many pages did it print?

Answer: _____ pages

Work space

14) Tom, Sam, and Kim have 54 coins total. Tom has 6 more than Sam; Sam has 6 more than Kim. How many coins does Tom have?

Answer: _____ coins

Work space

15) A bag has 4 yellow, 6 green, 5 orange candies. What is the fewest you must grab (without looking) to guarantee 3 of the same color?

Answer: _____ candies

Work space

16) A frog jumps 3 pads forward then 1 pad back each turn. Starting at pad 0, how many turns to reach pad 20?

Answer: _____ turns

Work space

17) Use digits 2, 3, 4, 5, 7 (each once) to make a 3-digit + 2-digit addition with the largest possible sum. What is that sum?

Answer: _____

Work space

18) 6 people each shake hands with every other person exactly once. How many handshakes in total?

Answer: _____ handshakes

Work space

19) A 'special' number is divisible by both 3 and 5. How many special numbers are there from 1 to 50?

Answer: _____ special numbers

Work space

20) 16 teams enter single-elimination (1 loss = out; every game has a winner). How many total games are played before a champion is crowned?

Answer: _____ games

Work space

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Answers & Solutions

1) Answer: 77

Solution: $156 - 79 = 77$.

2) Answer: 29 crayons

Solution: $3 \times 12 = 36$ crayons. $36 - 7 = 29$ crayons.

3) Answer: 23

Solution: 3, 7, 11, 15, 19, 23. The 6th term is 23.

4) Answer: 6 grapes

Solution: $45 \div 5 = 9$ each. $9 - 3 = 6$ grapes.

5) Answer: \$18

Solution: $3 \times \$4 = \12 . $\$30 - \$12 = \$18$.

6) Answer: 36 cm

Solution: $4 \times 9 = 36$ cm.

7) Answer: Wednesday

Solution: 20 days = 2 weeks + 6 days. Thursday + 6 = Wednesday.

8) Answer: 81 books

Solution: $120 - 35 + 18 - 22 = 81$ books.

9) Answer: 16 students

Solution: Piano: $30 \div 5 = 6$. Soccer: $30 \div 3 = 10$. Total: $6 + 10 = 16$.

10) Answer: 16

Solution: Work back: $47 - 15 = 32$; $32 \div 2 = 16$.

11) Answer: 12:25 PM

Solution: $9:45 + 2 \text{ h} = 11:45$; $11:45 + 40 \text{ min} = 12:25 \text{ PM}$.

12) Answer: 5

Solution: Known: $12+8+16+4 = 40$. Cole's pair = $50-40 = 10$; each card = 5.

13) Answer: 120 pages

Solution: $2:25-2:10 = 15 \text{ min}$. $8 \times 15 = 120$ pages.

14) Answer: 24 coins

Solution: $3x + 18 = 54 \rightarrow x = 12$. Tom = $12 + 12 = 24$ coins.

15) Answer: 7 candies

Solution: Worst case: $2+2+2 = 6$ (no trio). 7th pick guarantees 3 of one color.

16) Answer: 10 turns

Solution: Net +2 per full turn. After 9 turns: pad 18. Turn 10: +3 reaches pad 21 (passes 20). Answer: 10 turns.

17) Answer: 795

Solution: Largest 3-digit: 742; smallest 2-digit: 53. $742 + 53 = 795$.

18) Answer: 15 handshakes

Solution: $6 \times 5 \div 2 = 15$ handshakes.

19) Answer: 3 special numbers

Solution: Divisible by 15: 15, 30, 45. That is 3 numbers.

20) Answer: 15 games

Solution: Each game eliminates 1 team. $16 - 1 = 15$ games.

Quick Answer Reference

#	Answer	#	Answer	#	Answer	#	Answer	#	Answer
1	77	2	29 crayons	3	23	4	6 grapes	5	\$18
6	36 cm	7	Wednesday	8	81 books	9	16 students	10	16
11	12:25 PM	12	5	13	120 pages	14	24 coins	15	7 candies
16	10 turns	17	795	18	15 handshakes	19	3 special numbers	20	15 games

— The End —